

M.Tech CSE/DSC – Reinforcement Learning

Part A: Explore Reinforcement Learning Applications Identify and list any five real-world applications where Reinforcement Learning is used.

1. Autonomous Vehicles (Self-driving Cars)
2. Robotics and Industrial Automation
3. Healthcare and Personalized Treatment
4. Smart Traffic Signal Control
5. Recommendation Systems (Netflix, YouTube, Amazon)

Part B: Research Paper Exploration Select any two applications from Part A. For each application, identify one recent research paper (published within the last 3 years is preferred) and summarize it in the following format.

Item	Paper - 1	Paper - 2
Application	Autonomous Driving	Smart Traffic Signal Control
Paper Title	Reinforcement Learning for Autonomous Driving: Recent Advances and Future Challenges	Multi-Agent Reinforcement Learning for Traffic Signal Control: A Survey
Year	2024	2023
Authours	Vinod Sharma, Parwati Kumawat, Pankaj Kumar, S. Yaminipriya, Subbarao Chamarthi, Jitendra Kumar	Máté Kolat, Bálint Kövári, Tamás Bécsi and Szilárd Aradi
Journal / Conference	IEEE Transactions on Intelligent Transportation Systems	MPDI
Main Contribution	Reviews modern RL techniques for autonomous driving including lane keeping, decision making, obstacle avoidance, and discusses challenges such as safety, sample efficiency, and sim-to-real transfer.	Surveys recent Multi-Agent Reinforcement Learning (MARL) methods for adaptive traffic signal control, comparing cooperative and competitive learning strategies and identifying future research directions.
RL Algorithm	Deep Q-Network (DQN), Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC)	Multi-Agent PPO (MAPPO), Independent Q-Learning (IQL), Deep Q-Network (DQN), Advantage Actor-Critic (A2C)